Undertow - Step 1 complete: the data layer

Built on 2026-10-09 20:22:55 UTC from real Monad mainnet data, block 111,990,497. Progress report for the Metropolis build (deadline 13 October 2026).

What this step delivers

Undertow's warnings are only as good as the position data under them (the roadmap names this the biggest risk). Step 1 therefore does one thing: it reads every open position on Perpl, turns it into clean structured data, and proves the numbers match Perpl's own figures. Nothing visual is built yet; this is the foundation every later screen and the bot stand on.

Markets	Accounts with positions	Open positions	Resting orders	Open-interest checks
16 (12 live)	432	654	790	16 / 16 pass

What was built

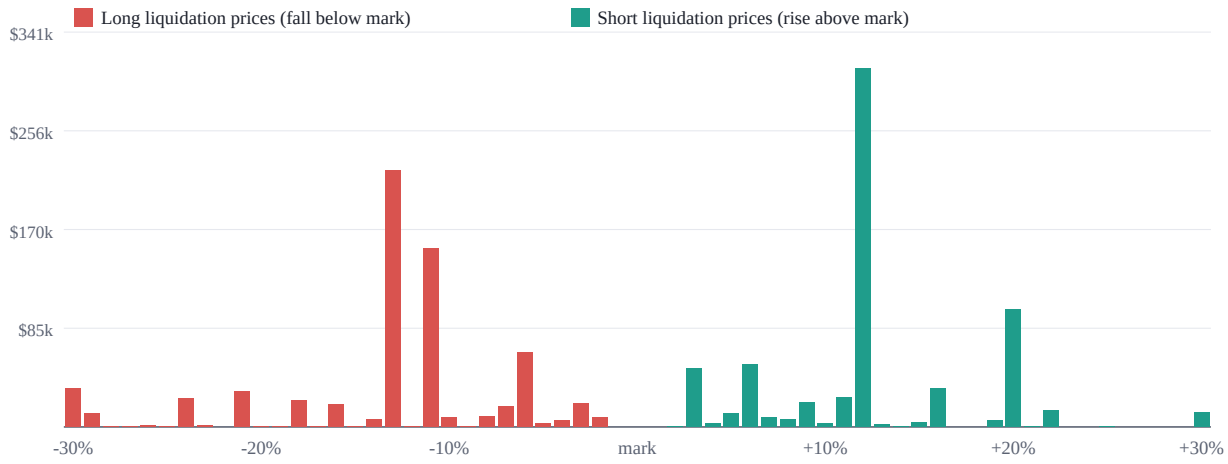
Piece	What it does
`undertow/perpl_cli.py`	Runs Perpl's official `perpl-cli` and captures its output as clean UTF-8. Read-only: the one command that signs transactions is deliberately not wrapped here.
`undertow/snapshot_parser.py`	Turns the snapshot text into typed records: markets (mark, oracle, last, funding, open interest, fees, margin settings), every resting order with its owner, and every position with Perpl's own liquidation and bankruptcy price.
`undertow/checks.py`	Correctness check 1 from the roadmap: rebuilt open interest must equal the figure Perpl reports.
`undertow/pipeline.py`	One command to fetch, parse, check, print a summary and save JSON (`data/parsed/snapshot_latest.json`).
`tests/test_parser.py`	5 automated tests against the real snapshot (block header, counts, a known account, BTC settings, OI check). All pass.

The important discovery

The roadmap planned to compute every trader's liquidation price ourselves. The snapshot already contains, for every position, **Perpl's own SDK-computed liquidation price and bankruptcy price**. So the exact liquidation data is available directly from the exchange's tooling, and our own formula (Step 3) becomes a cross-check and the engine for 'what if I used lower leverage' previews rather than the single source of truth. This removes the largest accuracy risk early.

First look at real data: where BTC traders get liquidated

Every BTC position, grouped by how far its liquidation price is from the mark price ($82,405.5), in 1% bins. Bar height is position value in USD. Values beyond 30% are shown in the 30% bin. This is a data preview, not the finished Radar.



Five biggest BTC liquidation clusters right now

Side	Liq. price vs mark	Position value	Positions
Short	+12%	$309,982	3
Long	-13%	$221,349	5
Long	-11%	$154,213	3
Short	+20%	$101,677	2
Long	-6%	$64,496	20

What it tells us

- Perpl is small, as the roadmap warned: BTC open interest is about $0.67 million. The largest single cluster is about $0.31 million. The pitch should say 'tools ready before Perpl grows', not 'millions at risk today'.
- The two largest clusters are held by only 3 and 5 positions. The biggest zones are a few large accounts, not a crowd. That fits the roadmap's point that a well-funded trader can see and target them.
- Close to the price (within 6%) the zones are thin ($50-65k each). The Radar and cascade simulator will show how little order-book depth sits against them.
- Sanity checks pass: no long has a liquidation price above the mark and no short has one below it; total long value equals total short value.

Correctness check 1: open interest (all markets)

Market	Rebuilt long size	Rebuilt short size	Perpl reports	Result
BTC	8.11273	8.11273	8.11273	PASS
MON	2.73927e+06	2.73927e+06	2.73927e+06	PASS
ETH	176.379	176.379	176.379	PASS
SOL_v2	1107.33	1107.33	1107.33	PASS
HYPE	439.98	439.98	439.98	PASS
ZEC	46.8751	46.8751	46.8751	PASS

Market	Rebuilt long size	Rebuilt short size	Perpl reports	Result
LIT	4942.5	4942.5	4942.5	PASS
VVV	357.43	357.43	357.43	PASS
PUMP	7.27546e+06	7.27546e+06	7.27546e+06	PASS
NEAR	1217.1	1217.1	1217.1	PASS
UNI	1049.93	1049.93	1049.93	PASS

Markets with no open positions (TAO and the four paused markets) are omitted from this table.

How to run it

From the project folder, in PowerShell:

```
python -m undertow.pipeline fetch
python -m undertow.pipeline parse data/snapshot.txt
python -m unittest discover -s tests
```

The snapshot is a full read of the exchange through Monad's public connection, which Perpl's tool limits to 15 requests per second. A full snapshot took a few minutes here. For a live dashboard Step 2 adds a faster, continuously updated feed and an Alchemy connection.

Honest limits of Step 1

- **Point in time only.** A snapshot is one block's state. Live updating (the 'trace' mode) is the next step.
- **Account IDs, not wallet addresses.** The snapshot labels traders by Perpl account number. The wallet explorer needs address-to-account mapping, which comes from `show account` or the Envio indexer.
- **Margin settings not yet interpreted.** BTC shows 'Margin: 15.00 / 25.00 (ini/mnt)'. I have not confirmed what these numbers mean in Perpl's formula, so Step 3 will verify our own liquidation formula against Perpl's liquidation prices before it is trusted.
- **Only checks 1 of 4 so far.** Open interest passes. 24-hour volume, our own test account and live order-book comparison come later.
- **Nothing trades yet.** No wallet, funds or orders are involved in this step.

What is left (against the roadmap)

Step	Scope	Status
2	Live feed: continuous updates, Alchemy connection, Envio indexer started for history	Next
3	Risk engine: liquidation formula checked against Perpl's, zones, cascade simulator	Not started
4	Radar map and sentence ('$42k of longs 2.1% below...')	Not started
5	Protocol view (volume, open interest, deposits, fees, funding, liquidations)	Not started
6	Wallet explorer, watchlist, compare, wallet statistics	Not started
7	Market-making bot: testnet first, then mainnet with hard limits and Telegram alerts	Not started - needs funded wallet
8	Order form with risk preview, funding cost tracker	Not started
9	Liquidation-aware orders, zone-aware bot	Not started
10	Remaining correctness checks, logo, README, GitHub share, four videos, submission	Not started

Schedule warning. Today is 10 October with the deadline on the 13th. The roadmap wanted the bot trading on mainnet from 9 October. That has not started and depends on funding a separate wallet, which needs you. I recommend starting the bot (testnet first) in parallel with Step 2, and deciding now which of the roadmap's 'cut in this order' items to drop.